

Statistical Modeling of Diamond Prices: An EDA-Guided Regression Analysis

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1 Introduction

Diamond prices are determined by a combination of physical and quality-based characteristics. Understanding how these features influence market prices is important for building accurate and interpretable pricing models. Such models have several practical applications in the gemstone industry, including,

1. Estimating the fair market value of diamonds and benchmarking listed prices.
2. Identifying potentially mispriced diamonds or pricing inconsistencies in inventory.
3. Assisting in the valuation of newly acquired batches of diamonds.

The objective of this work is to develop an interpretable statistical model for predicting diamond prices from their physical characteristics. This report presents details of a regression-based analysis of the `diamonds` dataset from the `ggplot2` R package, comprising observations of more than 50,000 diamonds with their corresponding market prices and various features. The study uses exploratory data analysis (EDA) as a guiding tool for feature engineering, and statistical regression to develop predictive models for estimating diamond prices.

The analysis also provides insight into the factors that influence diamond valuation.

2 The Dataset

The dataset consists of observations of 53940 diamonds. The observations include the following features :

1. **Price** of the diamond
2. **Carat** : Weight of the diamond
3. **Cut** : Diamond cut controls how much light reflects back, and hence the shine of the diamond. There are 5 categories. In ascending order of quality, these are : Fair < Good < Very Good < Premium < Ideal.
4. **Color** : Colorless diamonds are rare, since trace elements can turn the diamonds yellow/brownish. There are 7 categories, in ascending order of quality : J < I < H < G < F < E < D.
5. **Clarity** : Depending on whether the diamond displays internal/external flaws, there are 8 categories. In ascending order of rarity and quality, they are : I1 < SI2 < SI1 < VS2 < VS1 < VVS2 < VVS1 < IF.
6. **x,y,z** : Length, Width, Height of the diamond cut.
7. **depth** = $200 \text{ z}/(\text{x}+\text{y})$
8. **table** : width of the flat face expressed as a percentage of the diamond's widest part.

The correlation analysis of these features reveal **carat** to exhibit strongest association with **price**. The geometric features (**x**, **y**, and **z**) are also highly correlated with **price**, although this is mostly due to their strong correlation with **carat**. This suggests that the weight of the diamond is the dominant predictor of price. We find the remaining geometric variables to introduce substantial multicollinearity, without offering any improvement in predictive power.

3 Exploratory Data Analysis

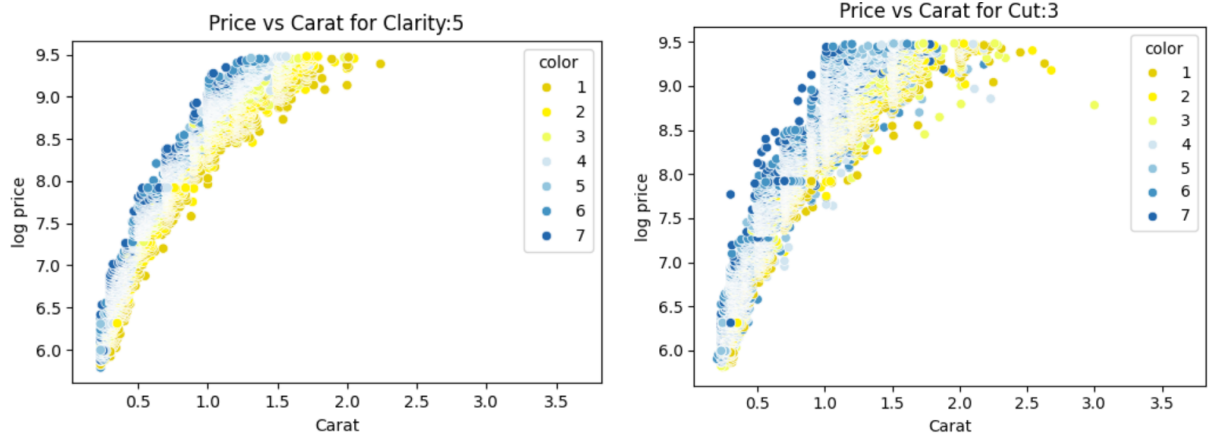
The response variable **price** spans several orders of magnitude. Thus, to handle heteroscedasticity, in our analyses we instead model $\log \text{price}$.

The dominant predictor is the diamond weight **carat**. Figure 1 shows the dependence of $\log \text{price}$ on **carat** for representative values of **clarity** and **cut**, with points colored according to **color**.

3.1 Dependence on carat

From Fig. 1, we observe

1. **Strong nonlinear dependence on weight** : $\log \text{price}$ increases rapidly with **carat** and the relationship is clearly nonlinear. This motivated the introduction of polynomial terms in **carat**, whose degree was later selected by cross-validation.
2. **Carat dominates all other predictors** : variations in diamond weight produce substantially larger changes in price than changes in **color**, **cut** or **clarity**. Consequently, the influence of these variables cannot be reliably assessed from the raw scatter plots alone.
3. **Evidence for interactions** : the band of the **color** groups appear to widen with **carat**, suggesting that the effect of **color** depends on weight, i.e. **carat** × **color** interaction terms are then motivated in the models.



(a) Scatterplot of `log price` v/s `carat` for a representative clarity (`clarity=5` corresponding to grade VS1). (b) Scatterplot of `log price` v/s `carat` for a representative cut (`cut=3` corresponding to grade Very Good).

Figure 1: Scatterplot of `log price` v/s `carat` . The data points are plotted with hue `color`

3.2 Residual Analysis

We remove a LOWESS fit of `log price` on `carat`, since the dominant dependence on `carat` masks the influence of the remaining predictors. The residuals were then examined, and we find,

1. **Interaction between colour and clarity** : Fig. 2 shows the median residual for each combination of `color` and `clarity`. Improvements in `color` produce larger increase in `price` for diamonds of higher `clarity`. This provides motivation for including a `color`×`clarity` interaction term in the regression model.



Figure 2: The median of residuals corresponding to every category of `color` and `clarity`.

2. **Positive dependence on clarity** : Fig. 3 illustrates the importance of accounting for confounding variables. The raw boxplots of `log price` appear to suggest that higher clarity corresponds to lower prices. However, this trend is entirely explained by the strong

correlation between clarity and carat. After removing the dominant carat dependence, the residuals exhibit the expected positive relationship between price and clarity.

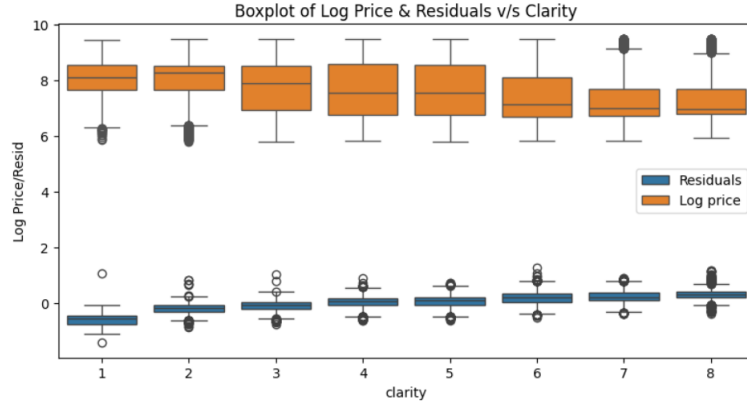


Figure 3: Boxplots of log price and residuals v/s clarity. The raw log price show a surprising downward trend, but once the effect of the confounding variable **carat** is taken into account, the (weak, but non-zero) positive dependence of the price is reflected in the boxplots of residuals.

3.3 Implications for feature engineering

The EDA motivates the following modeling decisions:

- Model the logarithm of price rather than the raw price.
- Represent the dependence on **carat** using polynomials.
- Include interaction terms involving **cut**, **color**, **clarity** and **carat**, among the candidate regression models.

4 Regression Models

The insights obtained from EDA are used for feature engineering. We proceed with model development by (i) determination of the functional dependence on **carat**, (ii) incorporating additional predictors and interaction terms, and (iii) validation of the final model using regularization, bootstrapping, and residual diagnostics.

4.1 Modeling carat dependence

Since EDA identified **carat** as the dominant predictor, we first determined the appropriate functional form for the dependence of $\log(\text{price})$. Polynomial models of degree one through four were compared using 15-fold cross-validation (CV). Fig. 4 shows the fold-wise mean validation MSE (mean squared error) together with its fold-wise standard deviation (shown as error bars). The one-standard-error rule on validation error selected the cubic model. Residual diagnostics, multicollinearity checks, and information criteria (AIC and BIC) also favored the cubic dependence.

We also try spline fits of **price** to **carat**. However the fit is unstable, provides poor interpretation power, and offers little to no improvement in prediction. Thus we proceed with the cubic **carat** model, and add other features to see whether the improvement in predictive performance is worth the complexity.

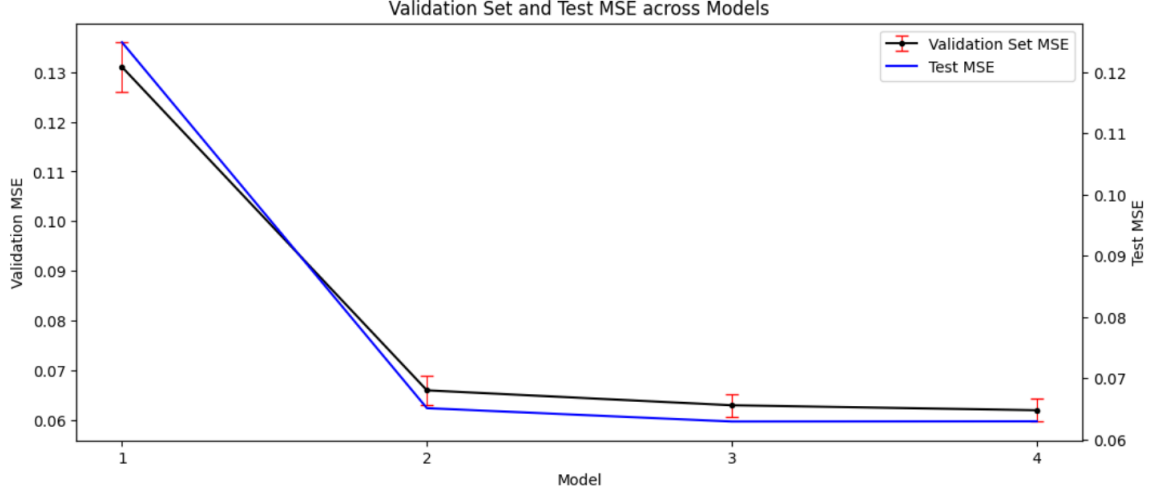


Figure 4: The validation and test set MSE across models. The test MSE is obtained from test set performance of the model trained on the full training dataset. Model n refers to the model $\log \text{price} = \beta_0 + \sum_{i=1}^n \beta_n \text{carat}^n$. Implementing the 1-SE (standard error) rule selects the cubic model as the best model.

4.2 Incorporating additional predictors

Adding the categorical variables `cut`, `color`, and `clarity` increased the test-set R^2 from 92.9% to 97.8%. Motivated by the interaction effects observed during EDA, candidate interaction terms were evaluated using cross-validation. The final model retained the `carat` $\times$ `clarity` and `clarity` $\times$ `color` interactions, yielding a further improvement to a test-set R^2 of 98.1%.

4.3 Model Diagnostics

We run a residual diagnostics on the model fit and find the distribution slightly skewed and tailed (skew = -0.135 , kurtosis = 5.572), but overall centered about 0, variance roughly constant across levels, and with no obvious structure left to exploit.

To assess the stability of the fitted model, approximately 50 bootstrap samples were generated. The resulting models exhibited extremely stable predictive performance, with a test-set $R^2 = 0.980875 \pm 1.25 \times 10^{-5}$. Furthermore, 99% of test observations exhibited less than 0.61% relative variation ($= \frac{\text{Std dev}}{\text{Mean}}$) in predicted prices across bootstrap samples.

4.4 Regularization and Automated Feature Selection

Ridge and LASSO regression were subsequently applied to investigate the stability of the selected feature set under coefficient shrinkage. Due to significant multicollinearity among polynomial and interaction terms, regularization redistributes explanatory power across correlated predictors while maintaining similar predictive performance. Importantly, the interaction terms `carat` $\times$ `clarity` and `clarity` $\times$ `color` remained non-zero over a wide range of penalty strengths, indicating that they continue to contribute predictive information even after regularization.

Finally, LASSO was fit using the complete engineered feature set, including variables excluded during the manual model-building process. Fig. 5 compares the validation performance of the regularized models with that of the manually selected OLS model.

The LASSO solution retained all predictors included in the manually engineered model, while shrinking several additional variables (for example, `carat` $\times$ `cut` and `y`) exactly to zero. The optimal penalty selected by cross-validation was $\lambda = 5.18 \times 10^{-5}$, yielding predictive performance nearly identical to the final OLS model. Thus, although LASSO explored a substantially richer

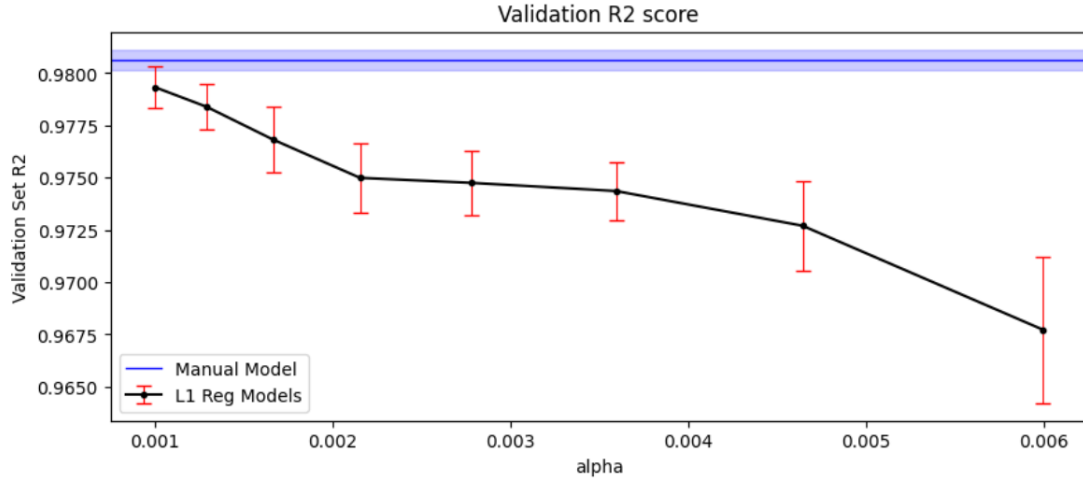


Figure 5: The validation set R^2 across L1 regularized models, with different penalty strengths α . The error bars denote fold-wise standard deviation of the R^2 scores. The validation set R^2 score of the unregularized model alongwith the fold-wise standard deviation is shown as a blue band.

feature space, it did not discover a meaningfully better predictive model than the simple and interpretable EDA-guided regression.

5 Principal Component Analysis

Principal Component Analysis (PCA) was performed to investigate whether dimensionality reduction could improve predictive performance. Inspection of the principal component loadings revealed that the first few components have clear physical interpretations:

- **PC1** is dominated by `carat`, `x`, `y`, and `z`, representing the overall *size* of the diamond.
- **PC2**, **PC3**, and **PC6** primarily combine `cut`, `depth`, and `table`, capturing the diamond’s *cut geometry*.
- **PC4** and **PC5** combine `color` and `clarity`, corresponding to the overall *quality* of the diamond.

Thus, PCA shows that the original nine predictors are largely organized around three latent factors: *size*, *cut geometry*, and *quality*.

5.1 Principal Component Regression

The principal component scores were subsequently used as predictors in a Principal Component Regression (PCR) model. Interestingly, the correlation of each principal component with the target price does not follow the ordering of explained variance. While the first principal component exhibits the strongest correlation with price, several lower-variance components remain predictive, indicating that directions explaining relatively little variance in the predictor space do contain predictive information.

Cross-validation showed that predictive performance improved monotonically as additional principal components were retained, with the best performance obtained using all principal components. Consequently, PCR offered no improvement over the regression models developed in the previous section. This behavior is expected for the present dataset. PCA only identifies directions of maximum variance without considering the response variable. Also, the dataset

contains a large number of observations relative to the number of predictors, so there is little variance reduction to be gained by discarding principal components. The cross-validation results therefore indicate that dimensionality reduction is unnecessary for this low-dimensional, high-sample-size dataset.

Overall, PCA proved valuable as an exploratory tool, identifying the dominant latent structure of the predictor space.

6 Conclusions

This work developed an interpretable statistical model for diamond price prediction through a combination of exploratory data analysis (EDA), feature engineering, cross-validation, and model diagnostics. We focussed on understanding the underlying pricing structure and ensuring that each modeling decision was well motivated.

EDA identified `carat` as the dominant predictor of price and motivated modeling the response using `logprice` to account for the large dynamic range of the raw prices. The analysis further revealed nonlinear dependence on `carat` together with weaker interaction effects involving the categorical quality variables. These observations directly guided the construction of the candidate regression models.

Cross-validation selected a cubic polynomial in `carat` as the optimal dependence. Incorporating the categorical variables `cut`, `color`, and `clarity`, together with the interaction terms `carat` $\times$ `clarity` and `clarity` $\times$ `color`, produced the best balance between predictive accuracy and model complexity, achieving a test-set R^2 of approximately 98.1%. Residual diagnostics and bootstrap resampling demonstrated that the fitted model is statistically well-behaved and highly stable under repeated sampling.

Regularized regression provided an independent assessment of the manually engineered feature set. Ridge and LASSO regression achieved predictive performance comparable to the final OLS model, while LASSO retained essentially the same set of important predictors despite searching a substantially richer feature space. This agreement indicates that the features identified through EDA capture the dominant sources of predictive information in the dataset.

Principal Component Analysis further revealed that the original variables are organized around three physically meaningful latent factors corresponding to diamond size, cut geometry, and quality. Although these components provide valuable insight into the correlation structure of the predictors, Principal Component Regression did not improve predictive performance, reflecting the relatively low dimensionality and high signal-to-noise ratio of the dataset.

Overall, in this project, through careful EDA-guided feature engineering and model validation, we produced an interpretable regression model whose predictive performance is comparable to that of more automated regularized approaches.

Future work could investigate nonlinear tree-based ensemble methods such as Gradient Boosting or XGBoost to determine whether additional predictive power can be obtained without sacrificing interpretability, as well as incorporating external variables describing market conditions.